SPADE · ENGINEERING WORKBENCH

Installation Guide

Step by step for Windows, Linux and macOS, written for everyone: from people who have never opened a terminal to people who want to modify the code.

Windows 10 / 11 | Ubuntu 22.04 · 24.04 · 26.04 | macOS Apple Silicon & Intel | No GPU needed

Two ways to install SPADE

Route A · Windows installer (.exe)

For people who just want to

use

SPADE on Windows.

Double-click and follow the wizard. No Python, no Git, no administrator rights. About 2 GB.

Route B · From the repository (source code)

For Linux and macOS, and for anyone who wants to

modify

SPADE or always run the latest version.

Install Python and Git, download the code and create a dedicated environment. About 3 GB.

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Code: **github.com/lucianoradrigan/SPADE** (public) · Installer: the repository's *Releases* page · Based on the README and `docs/INSTALL_*.md` · Screenshots: commit `7a05282`, release `app-v2` · October 2026

SECTION 1

Before you start

Five minutes of reading that prevent most installation errors.

1.1 Which route is right for me?

Are you on Windows and only want to use SPADE (not program or change the code)?

Yes → Route A (section 2): .exe installer | No → keep reading

Which operating system do you have?

Windows → section 3 | Linux (Ubuntu) → section 4 | Mac → section 5

	Route A · .exe installer	Route B · Repository
Systems	Windows 10 / 11, 64-bit only	Windows, Linux and macOS
What to install first	Nothing	Python 3.11 or 3.12 (64-bit) and Git
Administrator rights	No	On Windows, once (long paths). On Linux/Mac, to install system tools.
Difficulty	Very low	Medium: you type commands in a terminal
SPADE version	The one in the latest published installer (fixed)	The latest code; updated with one command
Can modify the code	No (not intended for that)	Yes

1.2 Computer requirements

Requirement	Details
Operating system	Windows 10 (21H2 or later) or 11, 64-bit · Ubuntu 22.04, 24.04 or 26.04 on an x86_64 processor · macOS on a Mac with Apple Silicon (M1–M4) or Intel
Disk space	About **3 GB** for Route B (mostly TensorFlow) · about **2 GB** for the .exe installer (≈290 MB download)
Graphics card (GPU)	**Not needed.** Everything runs on the processor. Messages about CUDA or GPU in the terminal are informational.
Processor	Must support AVX instructions (any ordinary PC from the last ten years). On very old or virtual machines without AVX, TensorFlow does not start.
Internet	Needed during installation (TensorFlow alone is ≈600 MB). Not needed to use SPADE afterwards.
Web browser	Any current one (Chrome, Edge, Firefox, Safari). SPADE is displayed in the browser at `http://localhost:8501`.

1.3 Basic concepts (if you have never used a terminal)

What is the terminal?

A window where you type instructions (commands) instead of clicking. Every command in this guide is copied, pasted and run with `Enter`.

- **Windows:** Start menu → type *PowerShell* → *Windows PowerShell*.
- **Ubuntu:** `Ctrl`+`Alt`+`T`, or search for *Terminal*.
- **Mac:** `⌘`+`Space` → type *Terminal*.

How to read the command blocks

```
python --version   # this is a comment
```

- Copy **only the left part**. Everything after `#` is an explanation and does not need to be typed (pasting it does no harm).
- Run the commands **in order**, one at a time, and wait for each to finish.
- Pasting into the terminal: `Ctrl`+`V` on Windows, `Ctrl`+`Shift`+`V` on Ubuntu, `⌘`+`V` on Mac.

What is a "virtual environment" (`.venv`)?

A folder inside SPADE with its own Python and libraries, separate from the rest of the computer. SPADE cannot break other programs and vice versa. When it is **activated**, the terminal line starts with `(.venv)`.

Passwords on Linux and Mac

Commands starting with `sudo` ask for your user password. **Nothing appears on screen while you type it** (not even asterisks): this is normal. Type it and press `Enter`.

SECTION 2 · ROUTE A

Windows installer (.exe)

The most direct way. No Python, no Git, no administrator rights.

Prerequisites

What you need	How to check / get it
Windows 10 or 11, **64-bit**	Start → Settings → System → *About* → "System type" must say *64-bit operating system*.
≈2 GB free disk space	File Explorer → *This PC* → check the free space on `C:`.
Internet connection for the download (≈290 MB)	—

Steps

1 Download the installer

Open https://github.com/lucianoradrigan/SPADE/releases and pick the newest release called **"SPADE for Windows app-v…"**, marked *Latest* (Figure 1). Scroll down to **Assets** and click `SPADE-Setup-<version>-<commit>.exe` (Figure 2). Do not download "Source code": that is Route B material.

✓ **You should see:** the .exe file in your Downloads folder.

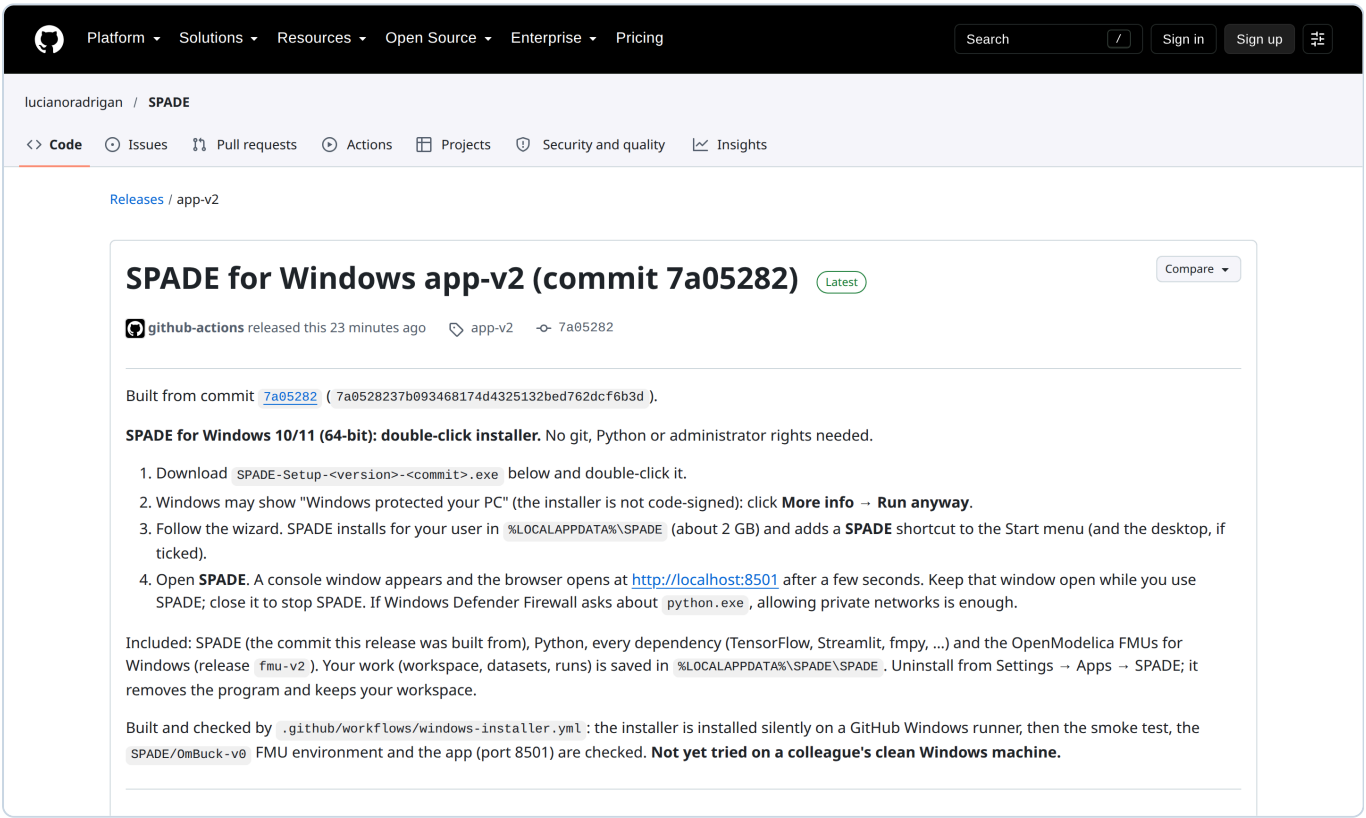


Figure 1. The release page on GitHub. The title says which commit the installer was built from; the green *Latest* label marks the newest version.



Figure 2. The *Assets* list at the bottom of the same page. Click the `.exe` (first line), not "Source code".

2 Double-click it

The installer is not digitally signed, so Windows may show **"Windows protected your PC"** (Microsoft Defender SmartScreen). This is expected: click **More info → Run anyway**.

✓ **You should see:** the SPADE setup wizard.

3 Follow the wizard

SPADE is installed only for your user in `%LOCALAPPDATA%\SPADE`. You can tick the box to create a desktop shortcut.

✓ **You should see:** a **SPADE** shortcut in the Start menu.

4 Open SPADE

Open **SPADE** from the Start menu (or desktop). A black console window appears and, after a few seconds, the browser opens.

✓ **You should see:** the SPADE *Workspace overview* page at `http://localhost:8501` (Figure 4, section 6).

Keep the black window open while you use SPADE. To close SPADE, close that window.

> **If Windows Firewall asks about `python.exe`**
> Allow access on **private networks**; that is enough. SPADE is only used from your own computer.

What is included and where your work is saved

- SPADE (the commit named in the file name), its own Python, every library (TensorFlow, Streamlit, fmpy…) and the OpenModelica FMU models for Windows.
- Your work (workspace, datasets, runs) is saved in `%LOCALAPPDATA%\SPADE\SPADE`.
- **Uninstall:** Settings → Apps → SPADE → Uninstall. The program is removed; your workspace is kept.
- **Update:** download and install the .exe of a newer release when there is one.

> **Verification status**
> The installer is built and tested automatically on a GitHub Windows machine (silent install, smoke test, FMU environment and the app on port 8501). It has **not yet been tried on a user's clean Windows PC**. The Windows dialogs in steps 2–4 are described but not pictured, because no Windows machine was available to capture them. If something fails, write down the message and report it.

SECTION 3 · ROUTE B

Windows 10 / 11 from the repository

Use **PowerShell** (not WSL). Open it from the Start menu: "Windows PowerShell".

Prerequisites and how to get them

Requirement	What it is for	How to get it
Python 3.12 (64-bit) 3.11 also works; 3.13 or newer does **not**	The language SPADE is written in.	**Option 1 (recommended):** `winget install --id Python.Python.3.12 -e` **Option 2:** python.org → Downloads → Windows → *Windows installer (64-bit)* for 3.12. On the first screen tick **"Add python.exe to PATH"**.
Git	Download the code and update it later.	**Option 1:** `winget install --id Git.Git -e` **Option 2:** git-scm.com → Download for Windows, accept the default options.
Long paths enabled	TensorFlow has very long file paths; without this the install fails.	One command as administrator (step 2).
Microsoft Visual C++ Redistributable (only if a DLL error appears)	Windows libraries used by TensorFlow.	`winget install Microsoft.VCRedist.2015+.x64`

> **No `winget`?**
> It ships with Windows 11 and updated Windows 10 (called *App Installer* in the Microsoft Store). If the command does not exist, use option 2 (download from the website) in each case.

Steps

1 Install Python 3.12 and Git

```
winget install --id Python.Python.3.12 -e
winget install --id Git.Git -e
```

Close and reopen PowerShell (so it finds the new programs), then check:

```
py -3.12 --version     # Python 3.12.x
git --version          # git version 2.x
```

2 Enable long paths (once)

Open PowerShell **as administrator** (right-click → *Run as administrator*), run the command below and **restart the computer**:

```
New-ItemProperty -Path "HKLM:\SYSTEM\CurrentControlSet\Control\FileSystem" -Name "LongPathsEnabled" -Value 1 -PropertyType DWORD -Force
```

✓ **You should see:** a line with `LongPathsEnabled : 1`.

3 Download the code into a short path

(normal PowerShell, not administrator)

```
mkdir C:\dev -Force
cd C:\dev
git clone https://github.com/lucianoradrigan/SPADE.git
cd SPADE
```

✓ **You should see:** the folder `C:\dev\SPADE` with the code.

4 Create the environment and install SPADE

(several minutes)

```
py -3.12 -m venv .venv
.\.venv\Scripts\Activate.ps1     # the line now starts with (.venv)
python -m pip install --upgrade pip
pip install -e ".[dev,viz]"
```

If activation fails with "*running scripts is disabled on this system*", run this once and activate again:

```
Set-ExecutionPolicy -Scope CurrentUser -ExecutionPolicy RemoteSigned
```

✓ **You should see:** at the end, `Successfully installed …` followed by a long list of packages.

5 Check the installation

```
python scripts\smoke_test.py
```

✓ **You should see:** every line with `[PASS]`; the last one `6. no exceptions: PASS` (Figure 3, section 6).

6 Open SPADE

```
streamlit run src\driveflow\viz\app.py
```

✓ **You should see:** the browser opens at `http://localhost:8501` with the Overview page (Figure 4). If the Firewall asks about `python.exe`, allowing **private networks** is enough. To stop SPADE, press `Ctrl`+`C` in PowerShell.

7 Next time

you only need:

```
cd C:\dev\SPADE
.\.venv\Scripts\Activate.ps1
streamlit run src\driveflow\viz\app.py
```

Verification status

These steps were checked by dependency resolution and the CI configuration; they have **not yet been run on a real Windows machine**.

SECTION 4 · ROUTE B

Linux (Ubuntu 22.04 · 24.04 · 26.04)

Open the Terminal with `Ctrl`+`Alt`+`T`.

Prerequisites and how to get them

Requirement	What it is for	How to get it
git, curl, build-essential	Download the code and compile some dependencies.	`sudo apt install -y git curl build-essential`
uv (recommended)	Installs a compatible Python by itself, without touching the system one.	`curl -LsSf https://astral.sh/uv/install.sh | sh`
Python 3.11 or 3.12	Run SPADE.	`uv` provides it automatically (step 4). Without uv: only Ubuntu 24.04's system Python works (see below).

Why uv? The Python that each Ubuntu ships

- **Ubuntu 22.04:** Python 3.10 → **too old**.
- **Ubuntu 24.04:** Python 3.12 → works.
- **Ubuntu 26.04:** Python 3.14 → **too new**.

With `uv` the commands are identical on all three. **Do not replace the system Python**: Ubuntu depends on it.

Steps (with uv, recommended)

1 Install the system tools

```
sudo apt update
sudo apt install -y git curl build-essential
```

2 Install uv

```
curl -LsSf https://astral.sh/uv/install.sh | sh
source "$HOME/.local/bin/env"
uv --version          # e.g. uv 0.9.x
```

3 Download the code

```
git clone https://github.com/lucianoradrigan/SPADE.git
cd SPADE
```

4 Create the environment and install SPADE

(several minutes; TensorFlow is ≈600 MB)

```
uv venv --python 3.11 .venv
source .venv/bin/activate     # the line now starts with (.venv)
uv pip install -e ".[dev,viz]"
```

5 Check the installation

```
python scripts/smoke_test.py
```

✓ **You should see:** every line with `[PASS]`; the last one `6. no exceptions: PASS` (Figure 3).

6 Open SPADE

```
streamlit run src/driveflow/viz/app.py
```

✓ **You should see:** the browser opens at `http://localhost:8501` (Figure 4). To stop SPADE: `Ctrl`+`C`.

7 Next time

```
cd SPADE
source .venv/bin/activate
streamlit run src/driveflow/viz/app.py
```

Alternative without uv (Ubuntu 24.04 only, with the system Python 3.12)

```
sudo apt install -y python3.12-venv
python3.12 -m venv .venv
source .venv/bin/activate
pip install -e ".[dev,viz]"
```

Exactly the verified environment

To get exactly the versions the test suite was verified with (Python 3.11.15, TensorFlow 2.21.0, Streamlit 1.62.0), also run `uv pip install -r requirements-lock.txt` in step 4. Linux is the system SPADE is developed and tested on.

SECTION 5 · ROUTE B

macOS (Apple Silicon & Intel)

Open the Terminal: ⌘ + Space → "Terminal".

Step 0 · Which Mac do I have?

```
uname -m      # arm64 = Apple Silicon (M1–M4) · x86_64 = Intel
```

Or: Apple menu (top left) → *About This Mac*: "Chip Apple M…" means Apple Silicon; "Processor Intel…" means Intel. If your Mac is Apple Silicon but `uname -m` says `x86_64`, the Terminal is running under Rosetta: use a native Terminal.

	Apple Silicon (M1–M4)	Intel
Python	3.11 or 3.12	**3.12 only**
Extra step	None	Install with the constraints file `constraints/macos-intel.txt` (step 4)
Why	—	Google no longer publishes new TensorFlow builds for Intel Macs (2.16 at most), which forces compatible versions of NumPy, pyarrow and Streamlit.

Prerequisites and how to get them

Requirement	What it is for	How to get it
Xcode Command Line Tools	Include Git and compilers.	`xcode-select --install` and accept the window that appears. If it says they are already installed, continue.
Homebrew	Package installer for Mac.	See step 2 (only if missing; check with `brew --version`).
Python 3.12 and Git	Run SPADE and download the code.	`brew install python@3.12 git`

Steps

1 Install the Command Line Tools

```
xcode-select --install
```

2 Install Homebrew

(only if `brew --version` fails), then Python and Git

```
/bin/bash -c "$(curl -fsSL https://raw.githubusercontent.com/Homebrew/install/HEAD/install.sh)"
brew install python@3.12 git
python3.12 --version     # Python 3.12.x
```

When it finishes, Homebrew may print "*Next steps*" commands to add it to the PATH: copy and run them.

3 Download the code

```
git clone https://github.com/lucianoradrigan/SPADE.git
cd SPADE
```

4 Create the environment and install SPADE

(pick the block for your Mac)
Apple Silicon:

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
pip install -e ".[dev,viz]"
```

Intel (do not skip `-c constraints/macos-intel.txt`):

```
python3.12 -m venv .venv
source .venv/bin/activate
python -m pip install --upgrade pip
pip install -e ".[dev,viz]" -c constraints/macos-intel.txt
```

5 Check the installation

```
python scripts/smoke_test.py
```

✓ **You should see:** every line with `[PASS]`; the last one `6. no exceptions: PASS` (Figure 3). The first run can be slow: macOS scans the new files.

6 Open SPADE

```
streamlit run src/driveflow/viz/app.py
```

✓ **You should see:** the browser opens at `http://localhost:8501` (Figure 4). If macOS asks whether Python may accept incoming connections, you can **Deny** for local use. To stop SPADE: `Ctrl`+`C`.

7 Next time:

`cd SPADE` → `source .venv/bin/activate` → `streamlit run src/driveflow/viz/app.py`

Do not install `tensorflow-metal`

SPADE does not need it and it was not tested with it.

Verification status

The macOS steps were checked by dependency resolution for both architectures and by emulating the Intel package set on Linux; they have **not yet been run on a real Mac**.

SECTION 6

What success looks like

Reference screenshots to compare with your own screen. Numbers such as versions and timings will differ slightly.

Smoke test (Route B, step 5)

```
(.venv) $ python scripts/smoke_test.py
SPADE smoke test -- Python 3.11.15 on Linux 7.0.0-38-generic (x86_64)
[PASS] 1. dependencies import (2.3 s) numpy 2.4.6; scipy 1.17.1; pandas 3.0.5; pyarrow 25.0.1; sklearn 1.9.
0; yaml 6.0.3; gymnasium 1.3.0; threadpoolctl 3.6.0; tensorflow 2.21.0; keras 3.15.1; streamlit 1.62.0; plot
ly 7.0.0
[PASS] 2. application starts (Overview page renders) (0.8 s) AppTest: no exceptions
[PASS] 3. platform modules load (0.0 s) rule sets: ['vsc_dpc']
[PASS] 4. short simulations (DC+PI, DC+MPC, VSC+DPC, PMSM FOC) (0.5 s) final speed 470.3 rpm, VSC |vc| 50.2
V
[PASS] 5. inference with a registered model (0.7 s) pc_server_5class_w2048/2026-10-20_run01 -> uncertain (0.
28, calibrated)
6. no exceptions: PASS -- 5/5 checks passed
```

Figure 3. Real output of the smoke test on Linux (TensorFlow's informational log lines removed). What matters: five `[PASS]` lines and `5/5 checks passed`. Any `[FAIL]` line names the part that broke; see section 8.

SPADE in the browser (all routes)

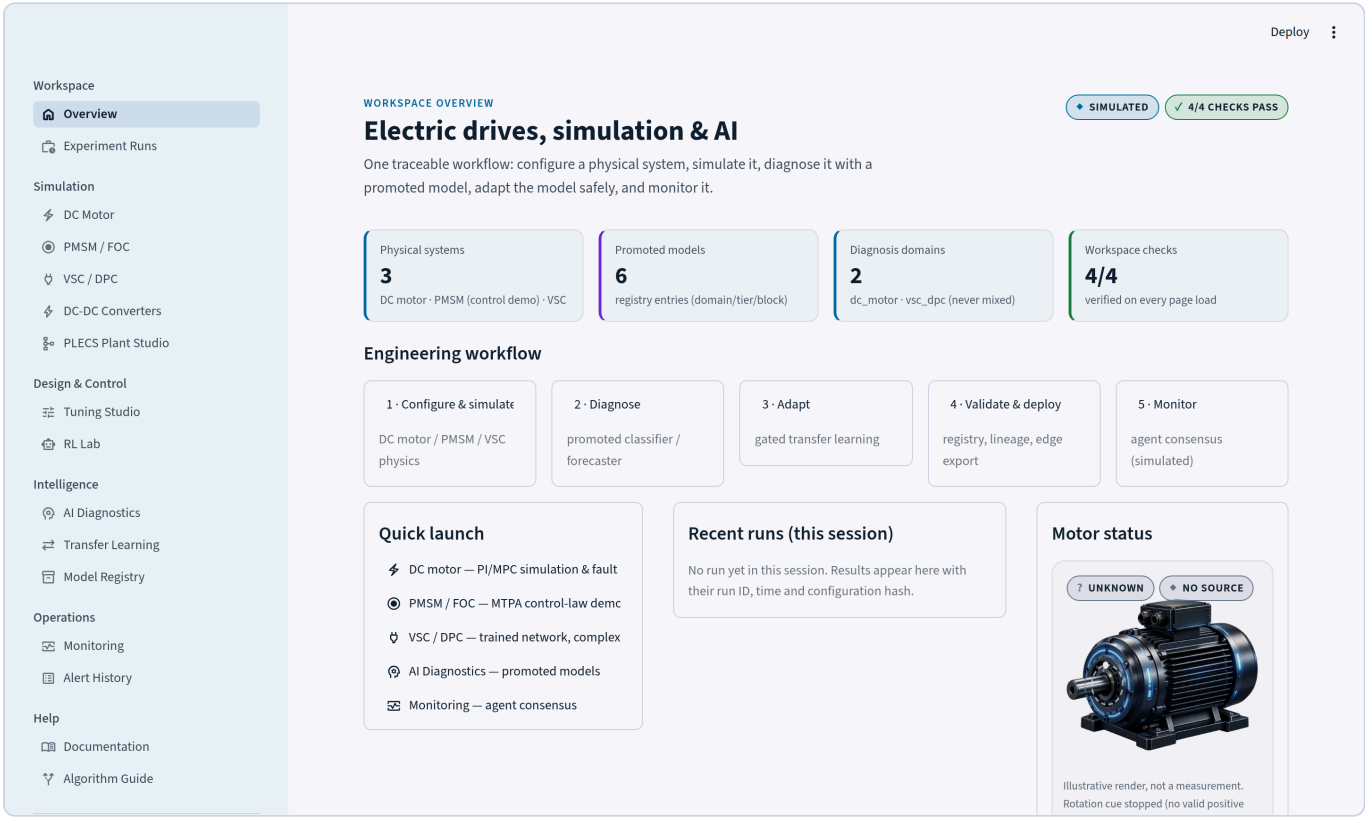


Figure 4. The page that opens at `http://localhost:8501`. Left: the navigation by group (Workspace, Simulation, Design & Control, Intelligence, Operations, Help). Top right: *4/4 CHECKS PASS* confirms the workspace is complete. "No run yet" and "UNKNOWN" motor status are normal on a fresh start.

SECTION 7

After installing

Daily use, updating and uninstalling (Route B; for Route A see section 2).

7.1 Opening SPADE every day

System	Commands
Windows	`cd C:\dev\SPADE` → `.\.venv\Scripts\Activate.ps1` → `streamlit run src\driveflow\viz\app.py`
Linux / Mac	`cd SPADE` → `source .venv/bin/activate` → `streamlit run src/driveflow/viz/app.py`

The legacy command `streamlit run src/driveflow/viz/dashboard.py` opens the same app and still works.

7.2 Long jobs (sweeps, tuning, RL, training)

They need the background **worker**. Two ways (Figure 5):

- In SPADE: **Experiment Runs → Start local worker**.
- Or in a second terminal, with the environment activated: `python -m driveflow.jobs.worker serve`

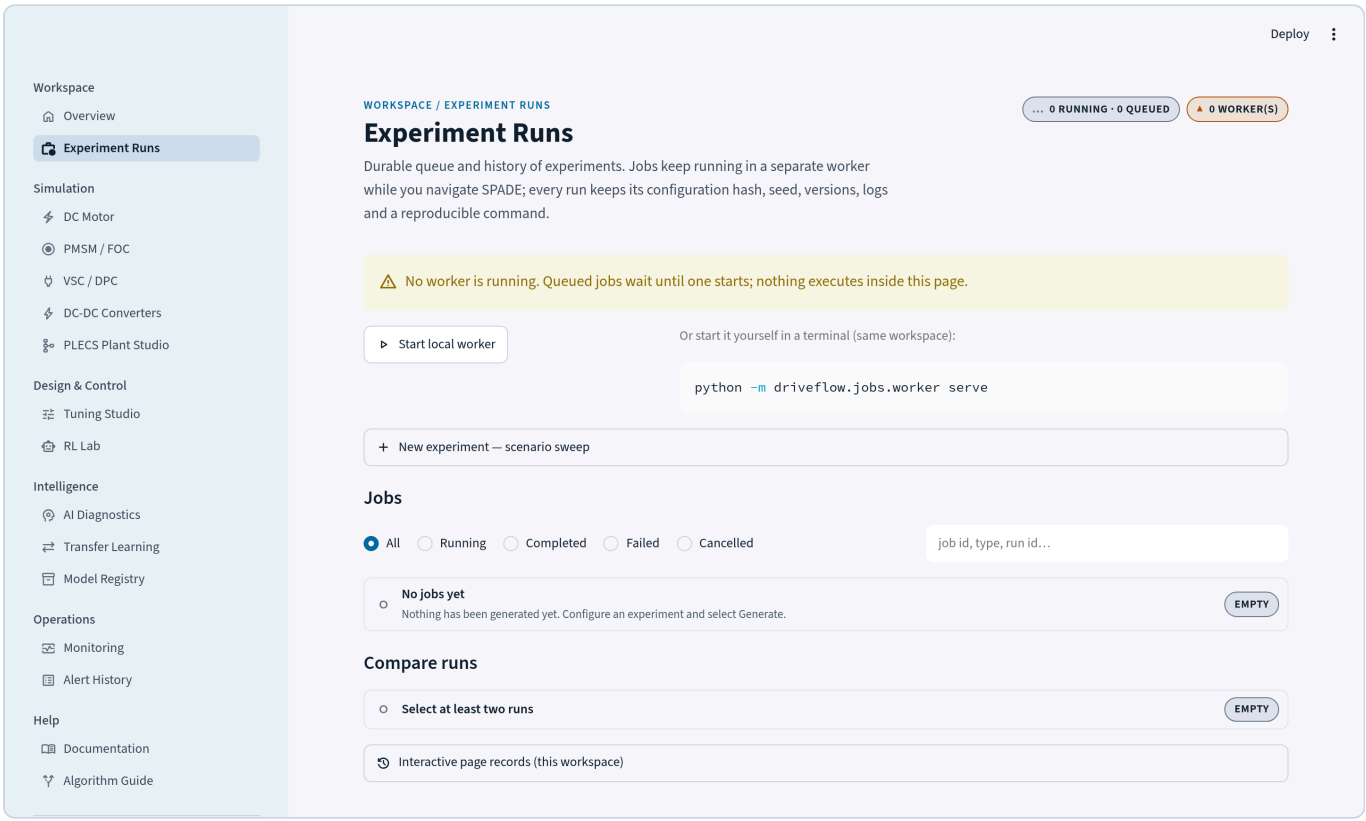


Figure 5. *Workspace → Experiment Runs* with no worker running: the yellow notice and the *0 WORKER(S)* badge say so. Click **Start local worker** or run the command shown on the right.

7.3 Updating to the latest version

With the environment activated, inside the SPADE folder:

```
git pull
pip install -e ".[dev,viz]"          # Linux with uv: uv pip install -e ".[dev,viz]"
# Intel Mac: add  -c constraints/macos-intel.txt
```

Then run `python scripts/smoke_test.py` again.

7.4 Uninstalling

- **Route B:** delete the `SPADE` folder (it contains the `.venv` environment). Keep or uninstall Python and Git like any other program.
- **Route A:** Settings → Apps → SPADE → Uninstall.

7.5 Useful settings

- **Light or dark theme:** ⋮ menu (top right) → Settings → Theme. By default it follows the system preference.
- **Another port** if 8501 is busy: `streamlit run src/driveflow/viz/app.py --server.port 8502` and open `http://localhost:8502`.
- **Hide TensorFlow's informational messages:** Linux/Mac `export TF_CPP_MIN_LOG_LEVEL=2` · Windows `$env:TF_CPP_MIN_LOG_LEVEL = "2"`.

> **About the data**
> Everything SPADE shows is **simulated**: there is no connection to real hardware. Pages say so with the *SIMULATED* label.

SECTION 8

Troubleshooting

Look for the message you see (even a partial match).

All systems

Message / symptom	Cause	Fix
`requires a different Python: 3.10…` / `3.13…` / `3.14…`	Unsupported Python version	Use Python 3.11 or 3.12 (Windows: `py -3.12`; Linux: `uv venv --python 3.11 .venv`; Mac: `python3.12`).
`streamlit: command not found` / `streamlit` not recognized	Environment not activated	Activate `.venv` (the line must start with `(.venv)`) or use `python -m streamlit run …`
`No module named 'streamlit'`	Installed without the GUI extra	`pip install -e ".[dev,viz]"` (with the quotes).
Messages about CUDA, cuDNN, GPU or oneDNN	Informational TensorFlow log	Ignore them: SPADE runs on the CPU.
`Illegal instruction (core dumped)` when importing TensorFlow	Processor without AVX (very old or virtual)	Use another computer, or a virtual machine with AVX enabled.
Port 8501 already in use	Another SPADE/Streamlit is open	Close it or use `--server.port 8502`.

Windows

Message / symptom	Cause	Fix
`running scripts is disabled on this system`	PowerShell execution policy	`Set-ExecutionPolicy -Scope CurrentUser RemoteSigned`
`WinError 206` / `No such file or directory …tensorflow\include…`	260-character path limit	Do step 2 (long paths), restart and clone into `C:\dev`.
`Could not find a version that satisfies the requirement tensorflow`	32-bit Python or 3.13+	Install 64-bit Python 3.12.
`DLL load failed while importing _pywrap_tensorflow_internal`	Missing Visual C++ Redistributable	`winget install Microsoft.VCRedist.2015+.x64` and reopen PowerShell.
`'python' is not recognized` or the Microsoft Store opens	Python not on the PATH	Use `py -3.12`, or turn off the "App execution aliases" for python in Settings.
"Windows protected your PC" with the .exe	Installer is not code-signed	More info → Run anyway.

Linux

Message / symptom	Cause	Fix
`ensurepip is not available`	The venv package is missing	`sudo apt install python3.12-venv` (or `python3.11-venv`).

MPC pages or tests very slow	Too many compute threads	`export OMP_NUM_THREADS=1`

macOS

Message / symptom	Cause	Fix
`pyarrow requires NumPy 2.0 or newer`	Intel Mac without constraints	Reinstall with `-c constraints/macos-intel.txt`.
`No matching distribution found for tensorflow==2.21`	Intel Mac (or Terminal under Rosetta)	Use the Intel constraints; on Apple Silicon, a native Terminal.
Errors mentioning `keras` 2.x	Python 3.11 on an Intel Mac	Recreate `.venv` with Python 3.12 and the constraints.
`illegal hardware instruction`	Environment created under Rosetta	Delete `.venv` and recreate it with `/opt/homebrew/bin/python3.12`.
`xcrun: error: invalid active developer path`	Command Line Tools missing	`xcode-select --install`

Nothing matches?

Copy the full error message (the last 20 lines of the terminal) and the output of `python scripts/smoke_test.py`, and include your operating system and the route you followed when reporting it. The full guides are in the repository's `docs/` folder.

APPENDIX

Glossary and quick summary

The technical words in this guide, one line each.

Term	Meaning
Repository (repo)	The project with all its source code, hosted on GitHub.
Git / clone	Git is the program that downloads and updates the code. "Cloning" means downloading a copy of the repository (`git clone`).
Python	The programming language SPADE is written in.
pip / uv	Programs that install Python libraries. `uv` is faster and can also install Python itself.
Virtual environment (.venv)	An isolated folder with SPADE's Python and libraries. It is "activated" before using SPADE.
winget / Homebrew / apt	Program installers for Windows, macOS and Ubuntu, respectively.
PATH	The list of folders where the system looks for the programs you type in the terminal.
Streamlit	The library that shows SPADE as a web page in your browser.
localhost:8501	Your own computer's address, on port 8501. Nothing goes out to the internet.
TensorFlow	The AI library used by SPADE's diagnosis models. The heaviest part of the installation.
Smoke test	A quick check (`scripts/smoke_test.py`) that confirms everything was installed correctly.
FMU	A packaged simulation model (FMI standard), here exported from OpenModelica.
Worker	A background process that runs SPADE's long jobs.

Summary table

	Windows · .exe	Windows · repo	Linux · repo	macOS · repo
Prerequisites	None	Python 3.12, Git, long paths	git, curl, build-essential, uv	Command Line Tools, Homebrew, Python 3.12
Install	Double-click the .exe	`pip install -e ".[dev,viz]"`	`uv pip install -e ".[dev,viz]"`	`pip install -e ".[dev,viz]"` (+ `-c constraints/macos-intel.txt` on Intel)
Check	Opens in the browser	`python scripts/smoke_test.py` → all `[PASS]`		
Open	Start menu → SPADE	`streamlit run src/driveflow/viz/app.py` → `http://localhost:8501`		
Tried on a real machine	GitHub runner (not a user's PC)	No (dependencies and CI only)	Yes (development system)	No (dependencies and emulation)